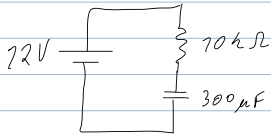


Exerzizium

1



$$Q = C \cdot \Delta V$$

$$t \rightarrow \infty \Rightarrow \Delta V_C = 12V$$

$$Q = 300\mu \times 12 = 3,6 \text{ mC}$$

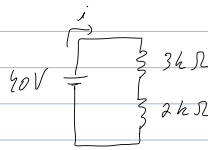
C

2

C

3

$$t \rightarrow \infty \Rightarrow i_C = 0 \quad \text{logos}$$

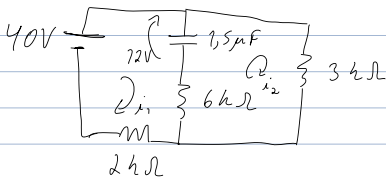


$$i = \frac{40}{3k + 2k} = 8 \text{ mA}$$

B

4

$$Q = 18\mu\text{C}, \quad \Delta V = \frac{Q}{C} = \frac{18\mu}{1,5\mu} = 12V$$



$$\begin{cases} -40 - 12 + 6k \cdot (i_1 - i_2) + 2k \cdot i_1 = 0 \\ 12 + 3k \cdot i_2 + 6k \cdot (i_2 - i_1) = 0 \end{cases} \Rightarrow$$

$$\Rightarrow \begin{cases} 8k \cdot i_1 - 6k \cdot i_2 = 52 \\ -6k \cdot i_1 + 9k \cdot i_2 = -12 \end{cases} \Rightarrow \begin{bmatrix} 8k & -6k \\ -6k & 9k \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 52 \\ -12 \end{bmatrix}$$

$$i_1 = \frac{\begin{vmatrix} 52 & -6k \\ -12 & 9k \end{vmatrix}}{\begin{vmatrix} 8k & -6k \\ -6k & 9k \end{vmatrix}} = \frac{52 \cdot 9k - (-6k) \cdot (-12)}{8k \cdot 9k - (-6k) \cdot (-6k)} = \frac{396k}{36k} = 11 \text{ mA}$$

C

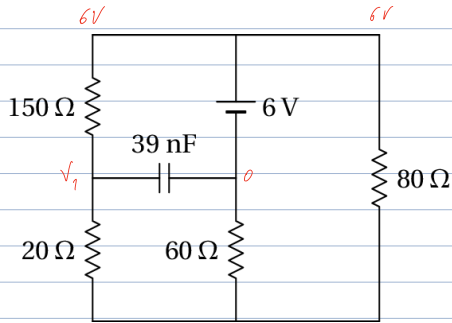
5

D

Problemlösung

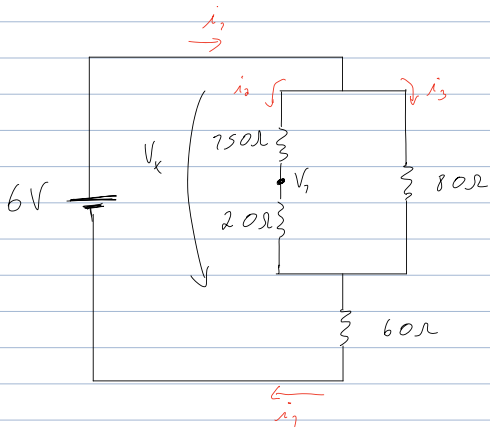
5.5 / 5.6

17



Energia final no Condensador?

(Circuito igual ao problema 6 mas com valores diferentes)



$$R_{eq} = ((150 + 20) // 80) + 60$$

$$= (170 // 80) + 60 =$$

$$= \frac{170 \times 80}{170 + 80} + 60 = 114,4 \Omega$$

$$i_1 = \frac{6}{114,4} = 52,4 \text{ mA}$$

$$V_x = 6 - 60 \times 52,4 \text{ mA} = 2,86 \text{ V}$$

$$6 - 11,8 - 1,150$$

$$i_2 = \frac{V_x}{150 + 20} = 16,8 \text{ mA}$$

$$=$$

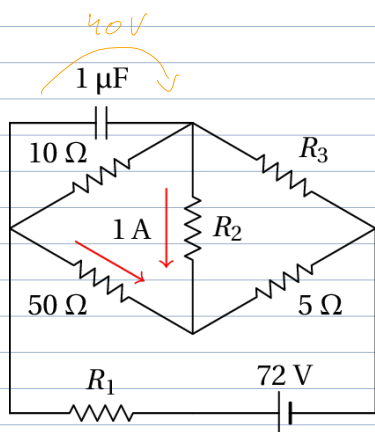
$$V_1 = V_{20\Omega} + V_{60\Omega} = i_2 \cdot 20 + i_1 \cdot 60 = 3,48 \text{ V}$$

$$Q_{\text{final}} = 39 \text{ nC} \times 3,48 = 136 \text{ nC}$$

Positiva em V1 e negativa em 0V

$$U = \frac{1}{2} \cdot Q \cdot V = \frac{1}{2} \cdot 136 \text{ nC} \times 3,48 = 237 \text{ nJ}$$

2



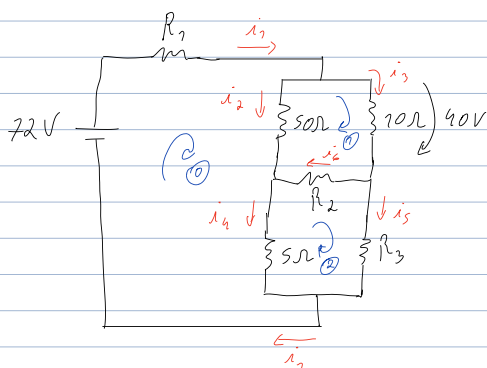
$$t \rightarrow \infty \Rightarrow Q_c = 40 \mu C$$

$$i_{50\Omega} = 1A$$

$$i_{R_2} = 1A$$

$$R_1, R_2, R_3 = ?$$

$$\Delta V_c \Big|_{t \rightarrow \infty} = \frac{Q_c}{C} = \frac{40 \mu}{1 \mu} = 40 V$$



$$i_6 = 1A$$

$$i_2 = 1A$$

$$i_3 = \frac{40}{10} = 4A$$

$$i_2 + i_6 = i_4 = 2A$$

$$i_5 + i_6 = i_3 \Leftrightarrow i_5 = 4 - 1 = 3A$$

$$i_7 = i_4 + i_5 = 5A$$

$$\Delta V_{R_1} = R_1 \times i_7 = 5 \cdot R_1$$

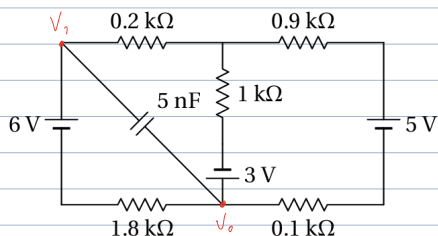
$$\Delta V_{50\Omega} = i_2 \times 50 = 50V$$

$$\textcircled{1} \quad 40 + R_2 \cdot i_6 - 50 \cdot i_2 = 0 \Leftrightarrow R_2 = \frac{-40 + 50}{1} = 10 \Omega$$

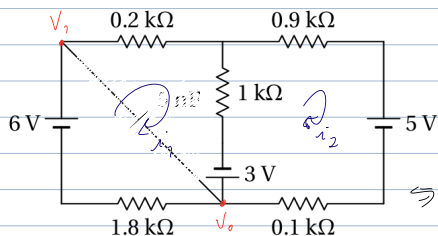
$$\textcircled{2} \quad R_3 \cdot i_5 - 5 \cdot i_4 - R_2 \cdot i_6 = 0 \Leftrightarrow 3 \cdot R_3 = 5 \times 2 + 10 \times 1 \Leftrightarrow R_3 = 6,67 \Omega$$

$$\textcircled{3} \quad -72 + R_1 \cdot i_7 + 50 \cdot i_2 + 5 \cdot i_4 = 0 \Leftrightarrow 5 \cdot R_1 = 72 - 50 \times 1 - 5 \times 2 \Leftrightarrow R_1 = 2,4 \Omega$$

3



$$Q_c \Big|_{t \rightarrow \infty} = ?$$



$$\begin{cases} -6 + 200 \cdot i_1 + 1k \cdot (i_1 - i_2) - 3 + 1,8k \cdot i_1 = 0 \\ 900 \cdot i_2 + 5 + 100 \cdot i_2 + 3 + 1k \cdot (i_2 - i_1) = 0 \end{cases} \Rightarrow$$

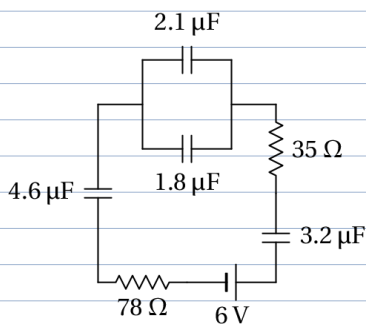
$$\Rightarrow \begin{cases} 3k \cdot i_1 - 1k \cdot i_2 = 9 \\ -1k \cdot i_1 + 2k \cdot i_2 = -8 \end{cases} \Leftrightarrow \begin{bmatrix} 3k & -1k \\ -1k & 2k \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 9 \\ -8 \end{bmatrix}$$

$$i_1 = \frac{\begin{vmatrix} 9 & -1k \\ -8 & 2k \end{vmatrix}}{\begin{vmatrix} 3k & -1k \\ -1k & 2k \end{vmatrix}} = \frac{9 \times 2k - (-1k) \cdot (-8)}{3k \cdot 2k - (-1k) \cdot (-1k)} = 2 \text{ mA}$$

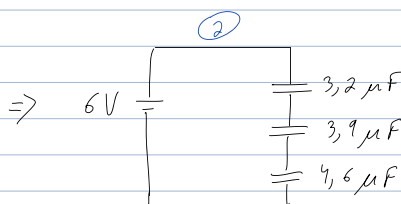
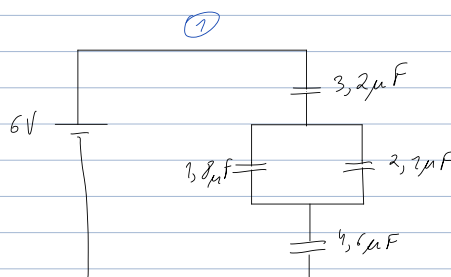
$$V_1 - V_0 = 6 - 1,8k \cdot i_1 = 2,4 \text{ V}$$

$$Q_c \Big|_{t \rightarrow \infty} = C \cdot \Delta V = 5n \times 2,4 = 12 \text{ nC}$$

4

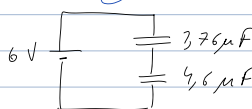


$t \rightarrow \infty$:

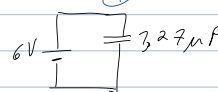


$\Rightarrow$

$$\frac{3,2\mu \times 3,9\mu}{3,2\mu + 3,9\mu} = 3,76\mu \quad ; \quad \frac{3,76\mu \times 4,6\mu}{3,76\mu + 4,6\mu} = 3,27\mu$$



$\Rightarrow$



$$⑦ Q_{1,27\mu F} = 6 \times 3,27\mu = 7,62\mu F$$

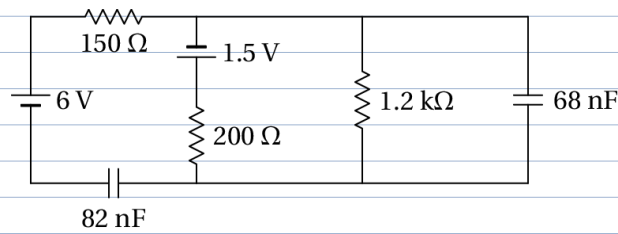
$$⑧ Q_{1,27\mu F} = Q_{3,2\mu F} = Q_{3,9\mu F} = Q_{4,6\mu F} = 7,62\mu F$$

$$\Delta V_{3,2\mu F} = \frac{7,62\mu}{3,2\mu} = 2,38V$$

$$\Delta V_{3,9\mu F} = \frac{7,62\mu}{3,9\mu} = 1,95V = \Delta V_{3,2\mu F} = \Delta V_{2,1\mu F}$$

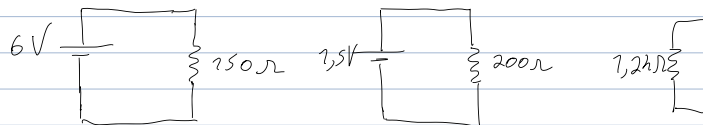
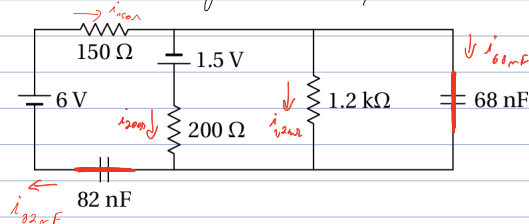
$$\Delta V_{4,6\mu F} = \frac{7,62\mu}{4,6\mu} = 1,66V$$

⑤ Condensadores inicialmente descarregados.



a) Correntes iniciais nas resistências e condensadores.

Condensadores descarregados comportam-se como curto-circuito.



$$i_{150\Omega} = \frac{6}{150} = 40mA$$

$$i_{200\Omega} = \frac{1,5}{200} = 7,5mA$$

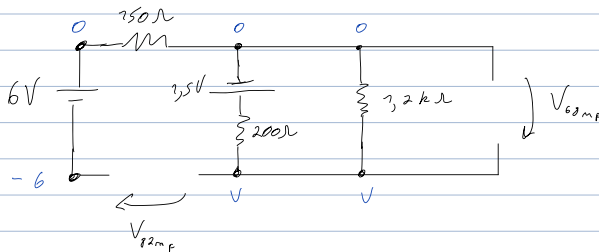
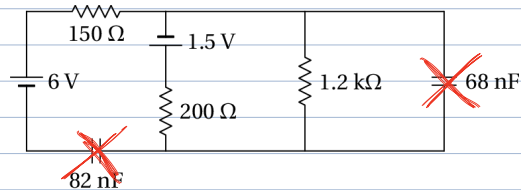
$$i_{1,2k\Omega} = 0mA$$

$$i_{150\Omega} = i_{200\Omega} + i_{1,2k\Omega} + i_{68nF} = i_{82nF} \Rightarrow$$

$$\Rightarrow 40 \text{ m} = 7,5 \text{ m} + 0 + i_{68 \text{ nF}} \Rightarrow \underline{i_{68 \text{ nF}} = 32,5 \text{ mA}}$$

le) Carregar pinos nos condensadores e polaridades.

Condensadores carregados comportam-se como circuito aberto:



$$V = ? \Rightarrow \begin{array}{c} \text{Circuit with } 1.5\text{V source, } 200\Omega \text{ resistor, and } 1.2\text{k}\Omega \text{ resistor.} \\ i = \frac{1,5}{1,2\text{k} + 200} = 1,07 \text{ mA} \\ V = 1,07 \text{ m} \cdot 1,2\text{k} = 1,284 \text{ V} \end{array}$$

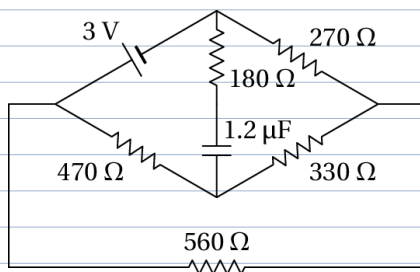
$$\begin{aligned} V_{82 \text{ nF}} &= 1,284 - (-6) = 7,284 \text{ V} & \text{(polaridade positiva no V e negativa em -6)} \\ V_{68 \text{ nF}} &= -1,284 \text{ V} & \text{(polaridade positiva no V e negativa no 0)} \end{aligned}$$

$$Q = C \cdot \Delta V$$

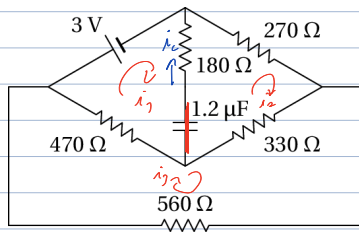
$$Q_{82 \text{ nF}} = 82 \text{ n} \cdot 7,284 = 597 \text{ nC}$$

$$Q_{68 \text{ n}} = 68 \text{ n} \cdot 1,284 = 87,3 \text{ nC}$$

6



a) Corrente inicial no condensador quando este descarregado.



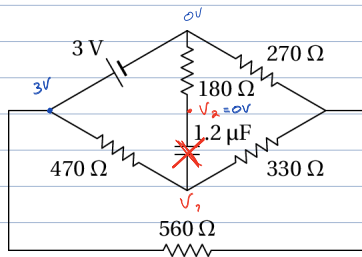
$$\begin{cases} 3 + 180 \cdot (i_1 - i_2) + 470 \cdot (i_1 - i_3) = 0 \\ 270 \cdot i_2 + 330 \cdot (i_2 - i_3) + 180 \cdot (i_2 - i_1) = 0 \\ 560 \cdot i_3 + 470 \cdot (i_3 - i_1) + 330 \cdot (i_3 - i_2) = 0 \end{cases} \Rightarrow$$

$$\begin{cases} 650 \cdot i_1 - 180 \cdot i_2 - 470 \cdot i_3 = -3 \\ -180 \cdot i_1 + 780 \cdot i_2 - 330 \cdot i_3 = 0 \\ -470 \cdot i_1 - 330 \cdot i_2 + 1360 \cdot i_3 = 0 \end{cases}$$

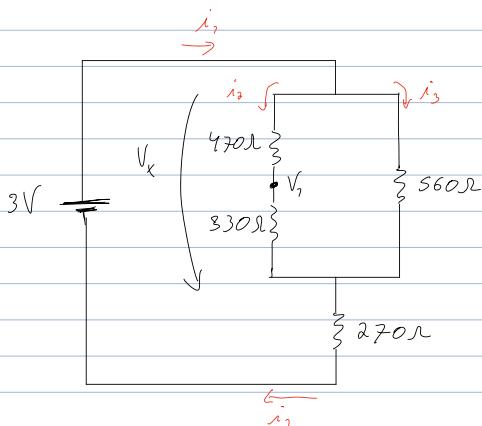
$$\begin{bmatrix} 650 & -180 & -470 \\ -180 & 780 & -330 \\ -470 & -330 & 1360 \end{bmatrix} \cdot \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} -3 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} -8,24 \\ -3,46 \\ -3,69 \end{bmatrix} \text{ mA}$$

$$i_c = i_2 - i_1 = -3,46 - (-8,24) = 4,78 \text{ mA}$$

b) Corrente do condensador quando carregado:



Como a resistência de 180Ω fica perdurando a queda de tensão é zero porque não passa corrente, logo $V_2 = 0 \text{ V}$.



$$\begin{aligned} R_{eq} &= ((470 + 330) // 560) + 270 \\ &= (800 // 560) + 270 = \\ &= \frac{800 \times 560}{800 + 560} + 270 = 599 \Omega \end{aligned}$$

$$i_1 = \frac{3}{599} = 5,01 \text{ mA}$$

$$V_x = 3 - 270 \times 5,01 \text{ m} = 1,65 \text{ V}$$

$$i_2 = \frac{V_x}{470 + 330} = 2,06 \text{ mA}$$

$$V_1 = V_{330\Omega} + V_{270\Omega} = i_2 \cdot 330 + i_1 \cdot 270 = 2,03 \text{ V}$$

$$Q_{1,2\mu} = 1,2 \mu \times 2,03 = 2,44 \mu \text{C} \quad \text{Positiva em } V_1 \text{ e negativa em } 0 \text{ V}$$